Interactively Visualizing Multivariate Market Segmentation Using the R Package Lionfish: Response to Review

We sincerely thank the editor and reviewers for your valuable suggestions, which have greatly improved the quality of both the article and the lionfish software.

General remark

The following summaries all of our changes and explanations to the comments provided. *Reviewer's comments are shown in italic.* When a comment resulted in specific changes in the text, we have added the original as well as the revised text to the response to that comment. Some additional changes were made to the paper, based on our own re-reading of the paper.

Editor

General Comments

The abstract and introduction should be improved to

(1) focus more on clustering results obtained using k-means clustering where a partition of the space is obtained which can be obtained using linear combinations of features,

(2) add more references to other visualization techniques for clustering solutions, data structure analysis in clustering and the importance of visualizations for market segmentation, in particular those by Fritz,

(3) add an overview on the structure of the manuscript at the end of the introduction where you outline the different sections and what they provide.

Specific Comments

- *Regarding (1), please note that one implicitly already focuses on such clustering results and some claims made would not hold for other clustering approaches used, e.g., auto-encoders, model-based approaches, etc. Later on only k-means clustering results are considered and this represents the default clustering method in many applications anyway. So it would ease the presentation to focus on this clustering method first and later on discuss extensions to other clustering approaches.*

  Response:

  Added text:

- *Regarding (2), please have a look at:*

  Nazila Babakhani, Friedrich Leisch, and Sara Dolnicar. A good graph is worth a thousand numbers. Annals of Tourism Research, 76:338-342, 2019. https://dx.doi.org/10.1016/j.a

  Friedrich Leisch. Neighborhood graphs, stripes and shadow plots for cluster visualization. Statistics and Computing, 20(4):457-469, 2010. https://dx.doi.org/10.1007/s11222-009-9137-8

  Friedrich Leisch. Visualizing cluster analysis and finite mixture models. In Chunhouh Chen, Wolfgang Härdle, and Antony Unwin, editors, Handbook of Data Visualization, Springer Handbooks of Computational Statistics, pages 561-588. Springer Verlag, 2008.

  Sara Dolnicar and Friedrich Leisch. Evaluation of structure and reproducibility of cluster solutions using the bootstrap. Marketing Letters, 21:83-101, 2010. https://dx.doi.org/10.10 009-9083-4

  Christian Hennig. Asymmetric Linear Dimension Reduction for Classification. Journal of Computational and Graphical Statistics, 13(4):930-945, 2004. https://doi.org/10.1198/

  Response:

  Added text:

- *Regarding (3), it might be good to emphasize the general applicability of the interactive tools provided and that the segmentation analysis is only one possible area of application which is, however, of focus in this contribution with the different illustrative applications. Note that Section 2 seems to be rather generic and not provide much information on use for inspecting clustering solutions, e.g., Section 2.1 does not mention any clustering results as input. So guiding readers on the different sections and what they present could help understanding.*

  Response:

  Added text:

Further Comments

- *Refer to the package being on CRAN including the URL (when first mentioning and in the Section "Resources and supplementary materials").*

  Response:

  Added text:

- *Please try to consistently refer to clusters / segments and avoid also using the term groups.*

  Response:

  Added text:

- *Please explain what "bit blit control" is as a reader might not be familiar with this and decide to write "bit blit" or "bit-blit".*

  Response:

  Added text:

- *It is unclear what is meant with "to numeric values that reflect the categories". Please be more specific.*

  Response:

  Added text:

- *On page 9, the extension to other feature types is unclear, e.g., how are feature means determined for ordered categories.*

  Response:

  Added text:

- *Regarding the importance of reproducibility one might also refer to other contributions in this SI:*

  Roger D. Peng. Tooling for Reproducible Research: Considerations for the Past and Future of Data Analysis. Austrian Journal of Statistics. doi:10.17713/ajs.v54i3.2052

  Robert Gentleman, Anthony Rossini and Vincent Carey: Contributions of Fritz Leisch to Vignettes and Reproducible Research. Austrian Journal of Statistics. doi:10.17713/ajs.v54i3.2057

  Response:

  Added text:

- *Note that Likert scales usually are referred to as multi-item scales where each item is evaluated on an ordinal scale ranging from disagree to agree and where an aggregate score across items is determined. Your data example hence thus not seem to include Likert data. You have ordinal/ordered categorical variables.*

  Response:

  Added text:

- *Not sure about "country's capital cities" as Austria clearly only has a single capital city or did you mean not country but federal state?*

  Response:

  Added text:

- *One might align the clusters in Figure 7 using classAgreement from package e1071 authored by Fritz Leisch.*

  Response:

  Added text:

- *The purpose of the paragraph "It is important to note that the silhouette scores ..." is rather unclear as well as its fit at this point in the manuscript.*

  Response:

  Added text:

- *The comment about removing duplicates in the risk data set is unclear. Given that the data consists of answers on 6 variables with 5 categories, it is unclear why respondents who give the exact same answers are to be removed. It would seem that these duplicates still represent genuine observations.*

  Response:

  Added text:

Stylistic issues

- *Please consistently use either British or American English.*

  Response:

  Added text:

- *Please use consistent names for the data set (including consistent lower/upper case). Also the dash in "Risk-taking" is unclear.*

  Response:

  Added text:

- *Omit numbers from Sections 7 and 8.*

  Response:

  Added text:

- *Use pkg mark-up for packages as defined in ajs.cls.*

  Response:

  Added text:

- *Please use "datapoints" or "data points".*

  Response:

  Added text:

- *Check when abbreviations (e.g., GUI) are introduced the first time and then consistently use them.*

  Response:

  Added text:

- *Use suitable markup for classes and indicate character strings correctly.*

  Response:

  Added text:

- *Use consistent markup for file extensions.*

  Response:

  Added text:

- *Figure missing in "as seen in 13".*

  Response:

  Added text:

- *Use a consistent labeling for "Guided tour - LDA - regroup".*

  Response:

  Added text: